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Human-to-Robot Transfer via Bridging Representation and Action Space in VLAs

H2RB: Unpaired Human Hand Videos Teach Pretrained VLAs New Objects and Skills

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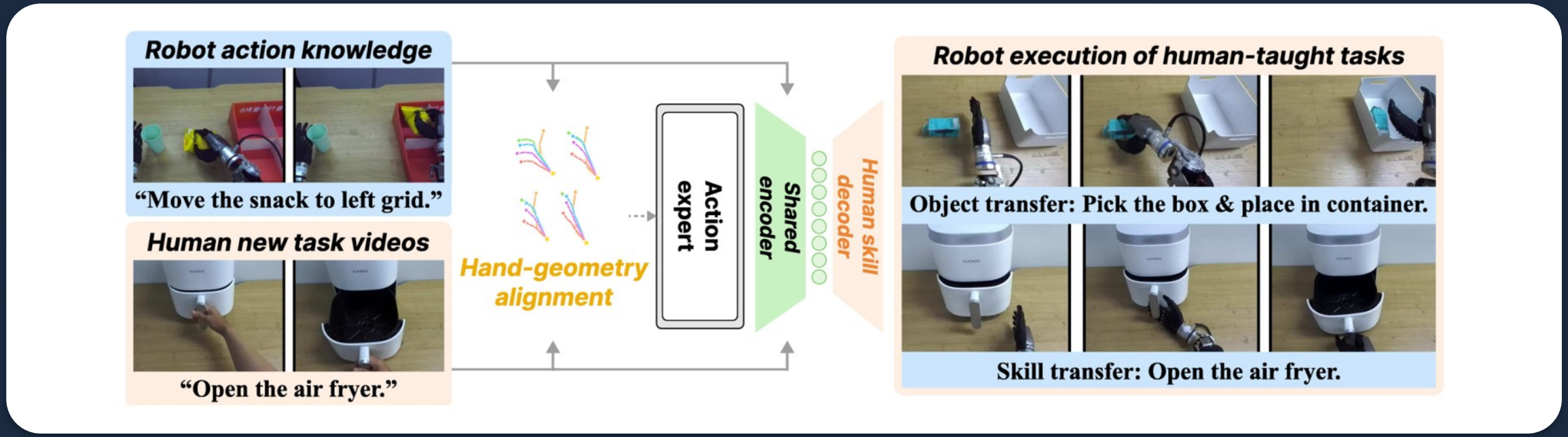


Figure: paper Figure 1 (H2RB human-to-robot task transfer)

Key Findings

Bridging perception and action makes human-to-robot transfer work

H2RB builds on a robot-pretrained VLA, aligning VLM features to wrist + five-fingertip geometry (perception) and using a latent action tokenizer with a shared encoder and separate human/robot decoders (action), so a real robot learns new objects and skills from unpaired human videos alone.

8.0 → 51.8

Object transfer score

Vanilla co-training → H2RB
OpenArm; Box/Ball/Towel: human demos only

0 → 62.7

Skill transfer score

Air fryer open/close: shared-action baselines all 0
Skill absent from robot data

21.3 → 45.3

OOD object transfer

Ego-Pi shared action → H2RB
Objects outside training quadrants

With large-scale heterogeneous human–robot training: object 57.2 / skill 81.7; RB-Y1 robot-only adaptation success 64% → 78%. (scores: 0–100, progress-weighted)

Problem & Motivation

Unifying action coordinates is not enough: two more gaps remain

① Action-space gap

Latent actions (inferred from frame changes) or reconstructed world-frame hand motion (HaWoR, Ego-Pi) place humans and robots in one action space.

✓ Partly solved

Addressed by prior work

② Perception gap

Human hands in one view, robot hands in the other; VLM features lie far apart, so the model must map very different observations to the same action.

✗ Unsolved by shared actions

H2RB: hand-geometry-anchored alignment

③ Execution gap

Same parameterization, different dynamics: humans move faster with different trajectory structure; forcing one decoder mapping causes interference.

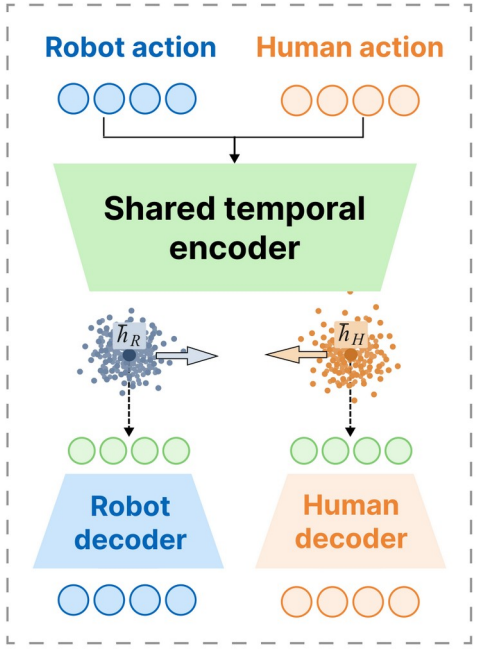
✗ Unsolved by shared actions

H2RB: latent action tokenizer with separate decoders

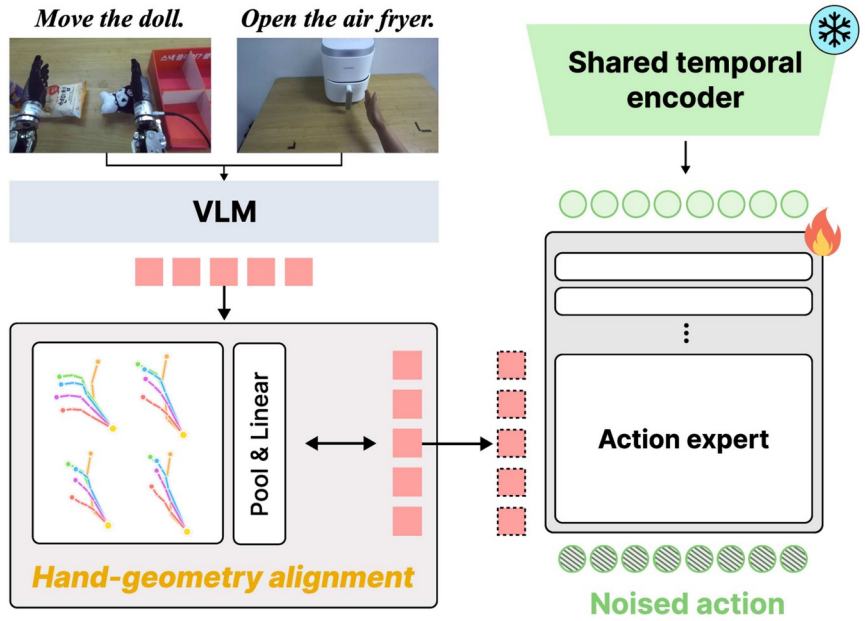
Setting: start from a robot-pretrained VLA; human and robot demos are unpaired, with no frame-level correspondence; target tasks have only human demos, no target robot demos.

Method Overview

H2RB: shared latent actions + hand-geometry-anchored perception



(a) Decoder-separated latent action tokenizer



(b) Hand-geometry-anchored VLM conditioning alignment + latent flow matching

Stage 1
Train tokenizer (100k steps)
Then freeze encoder and both decoders

›

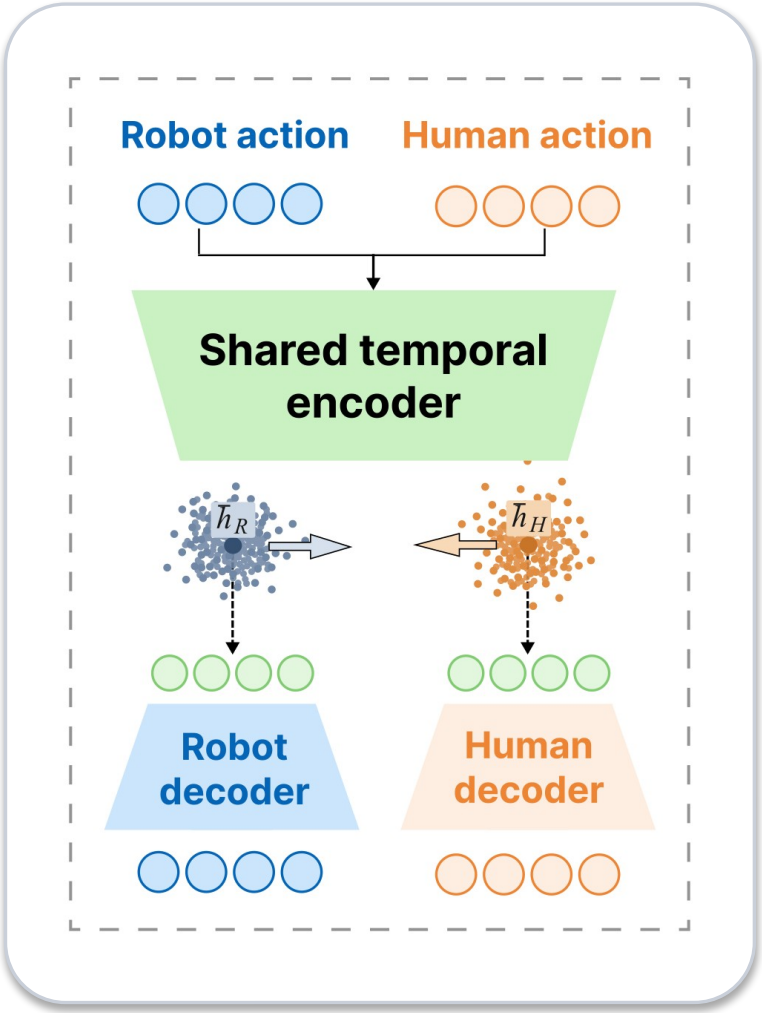
Stage 2
VLA fine-tuning: L = L_pred + 0.2 • L_geo
VLM backbone and tokenizer frozen

›

Deploy
Latent actions → human decoder D_H
→ IK: wrist pose to arm commands

Method ① Decoder-separated latent action tokenizer

Action side: shared encoder = “what”, separate decoders = “how”



Unified parameterization

Human hands reconstructed with HaWoR and retargeted; action = 3D wrist position + 6D rotation + hand joints (15-D per OpenArm hand)

Shared encoder + two full decoders

VAE: 16-step action chunk → 16×64 latents; each decoder has its own Transformer blocks and output head, not just a separate final layer

Cross-domain latent alignment

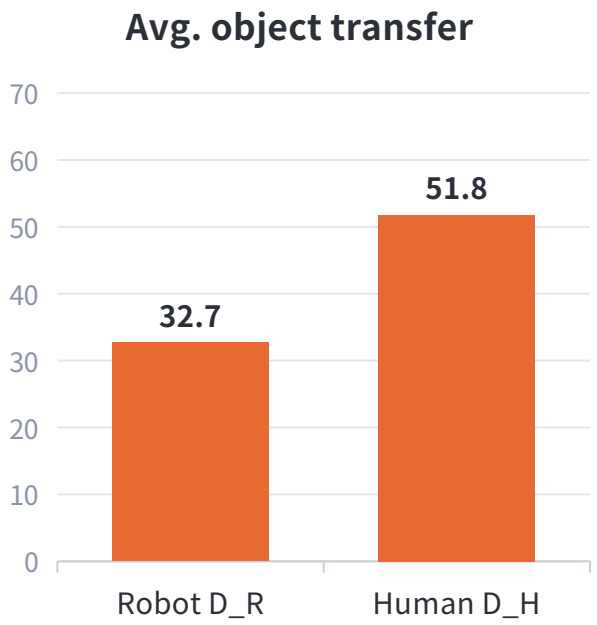
Centroid matching per chunk position, plus variance/covariance regularization against collapse

Frozen, then used as prediction target

The VLA performs flow matching on the posterior

L_tok = L_rec + 1e-6 • L_KL + 0.03 • (L_cent + L_var + 0.004 • L_cov)

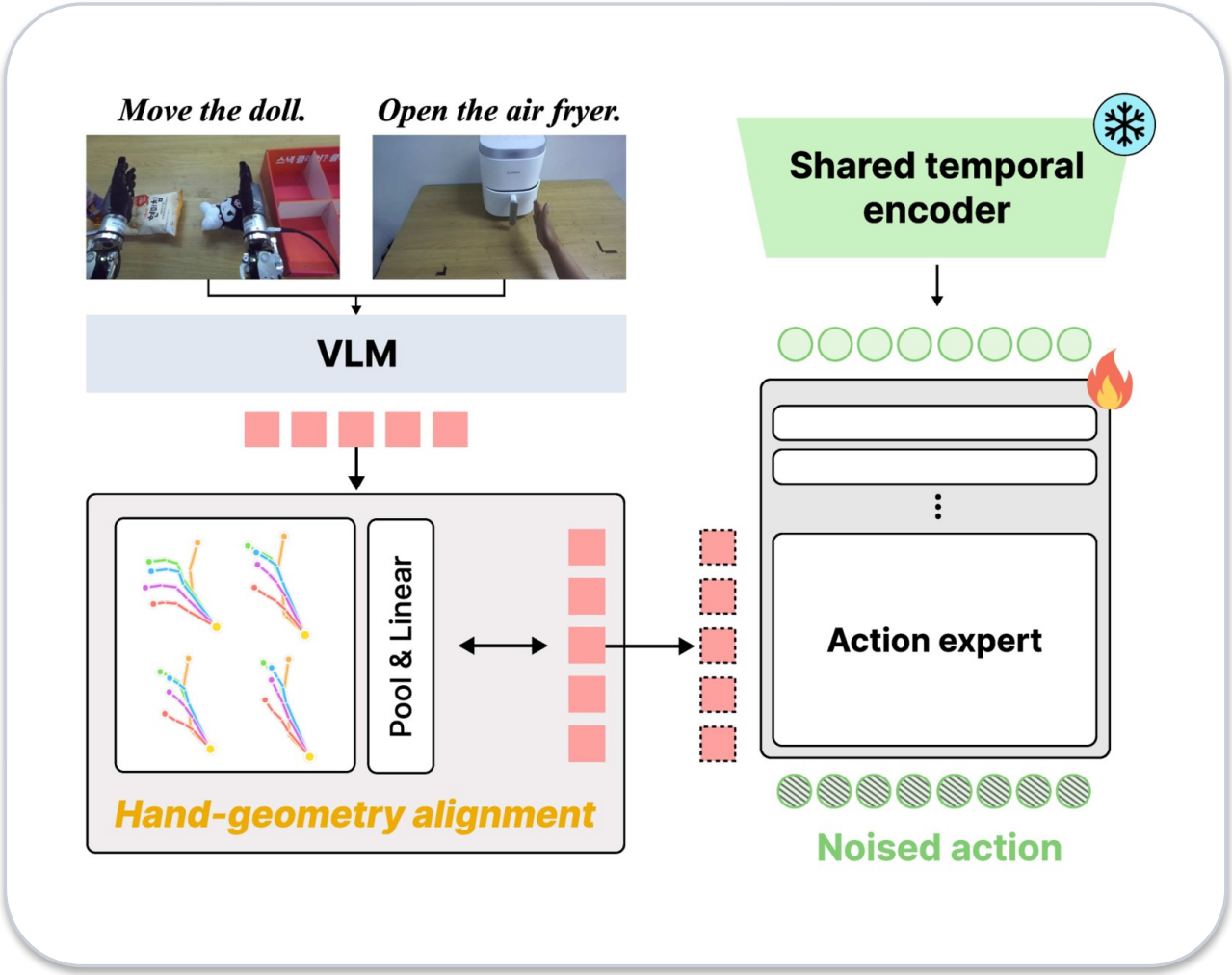
Counterintuitive: robot uses the human decoder



Same latent policy, only the decoder swapped (Table 13). The human decoder preserves the target behavior in human demos

Source: Figure 2(a), Section 2.2, Appendix A.2, Table 13

Perception: per-frame hand geometry as anchors, no paired data



Geometry target g_t (24-D per hand)

Current wrist pose + 5 fingertip positions in the wrist frame; from reconstruction (human) or forward kinematics (robot)

Residual adapter

$h_{geo} = h + A_\phi(h)$: token-wise MLP (width 256), zero-initialized output layer; the action expert conditions on h_{geo}

Training-only readout

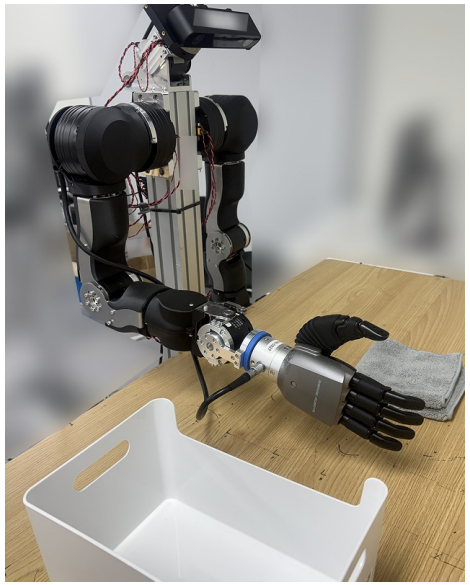
Mean-pool h_{geo}, then linearly regress g_t; masked Smooth-L1, $\lambda_{geo} = 0.2$

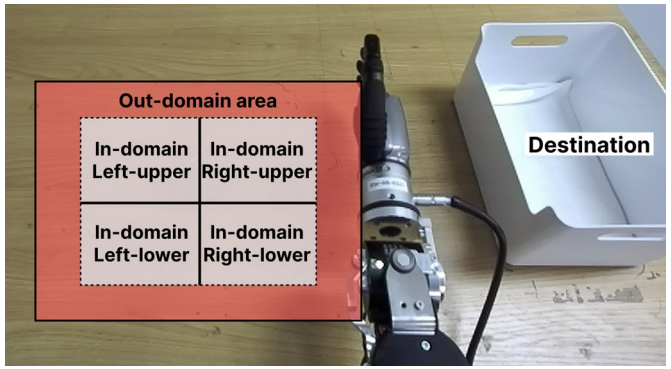
Why it scales

Every frame is self-labeled → no human–robot pairing, temporal correspondence or image synthesis (vs. PHANTOM / Masquerade)

Experimental Setup

Target tasks have only human demos: can the real robot learn them?





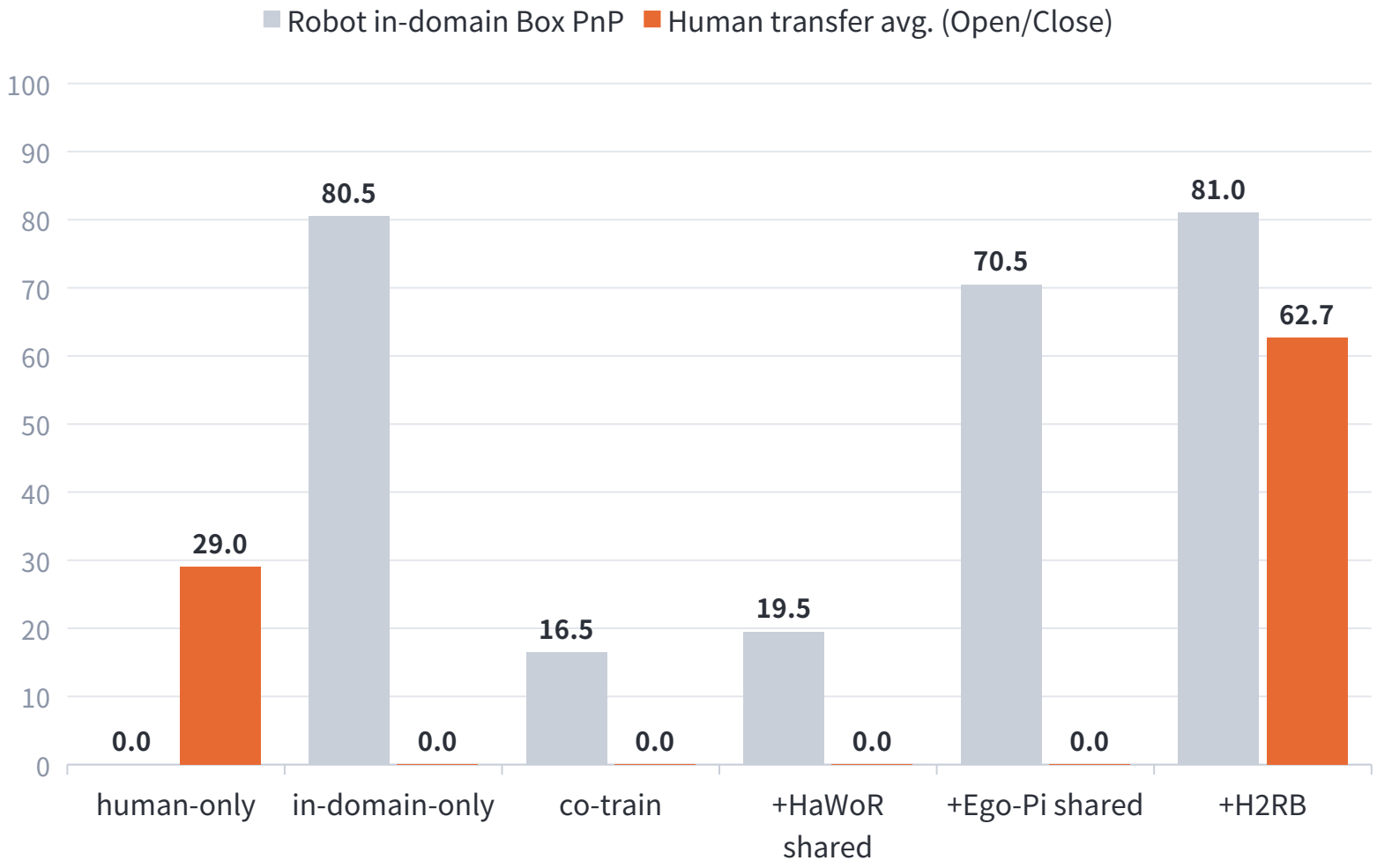
	Object transfer	Skill transfer
Human target tasks	Box/Ball/Towel PnP	Air fryer open/close
Human demos	100 per object	50 per task
Robot data	208 PnP, 4 objects	300 PnP, 3 objects
Robot in-domain eval	Doll PnP	Box PnP
Transfer eval	20 ID + 10 OOD per object	15 each Open/Close

Left: OpenArm + Inspire dexterous hands, head-mounted ZED 2i, 20 Hz; right: central four quadrants = ID placement, outer red zone = OOD

Init: all policies start from RLDX-1-IMG. **Metric:** progress-weighted score 0–100 (0 fail / 0.3 reach / 0.6 grasp / 0.8 lift or half open/close / 1.0 complete). **Baselines:** human-only, in-domain-only, co-train, HaWoR / Ego-Pi shared action, with equal data and training budget.

Main Results ①

Skill transfer: shared-action 0 → H2RB 62.7, no in-domain loss



Open 50.7 • Close 74.7

H2RB scores on the two new skills

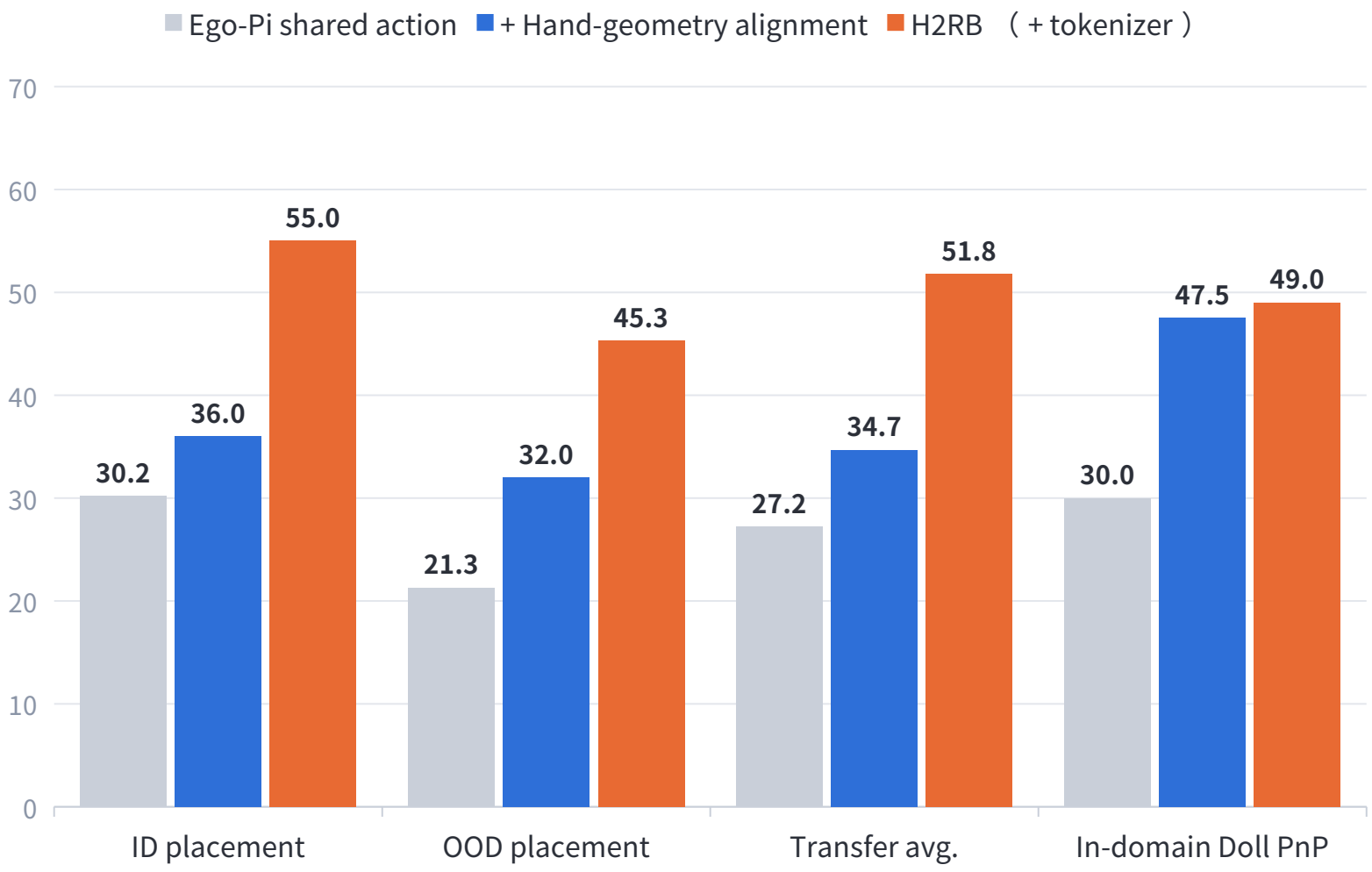
Observations

- Shared action spaces recover in-domain skill (Ego-Pi 70.5), but new skills stay at 0
- Human-only data learns motion patterns (29.0) but lacks robot execution knowledge
- H2RB gets both: in-domain 81.0 + transfer 62.7

Source: Table 1 (skill transfer)

Main Results ②

Object transfer: alignment and tokenizer gains stack, and hold OOD



8.0 → 51.8

Vanilla co-training → H2RB (main text)

Robot-only data: 12.0

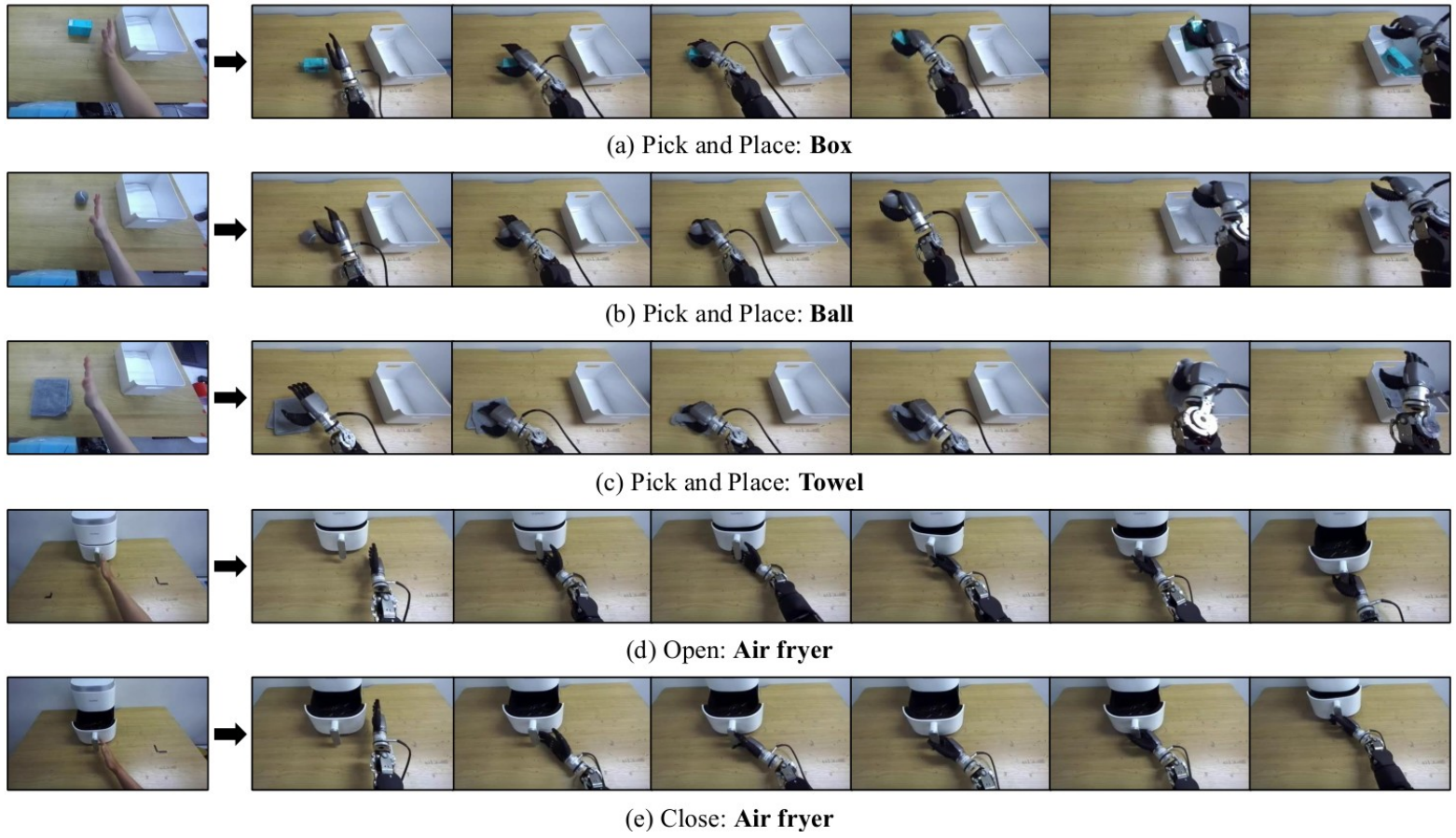
Observations

- Vanilla co-training is worse than robot-only data (8.0 < 12.0)
- Geometry alignment lifts the average from 27.2 to 34.7; the tokenizer, to 51.8
- OOD rises from 21.3 to 45.3 (ID: 55.0), retaining most gains out of distribution

Source: Section 3.2, Table 12

Main Results ③

Qualitative: from human demos, the robot completes five new tasks


(a) Pick and Place: **Box**

(b) Pick and Place: **Ball**

(c) Pick and Place: **Towel**

(d) Open: **Air fryer**

(e) Close: **Air fryer**

How to read this figure

- Leftmost in each row: a frame of the human demo
- Right of the arrow: successful H2RB execution on OpenArm

Two kinds of transfer

- Object-level: Box / Ball / Towel PnP (towel is deformable)
- Skill-level: opening/closing an air fryer (never in robot data)

Note

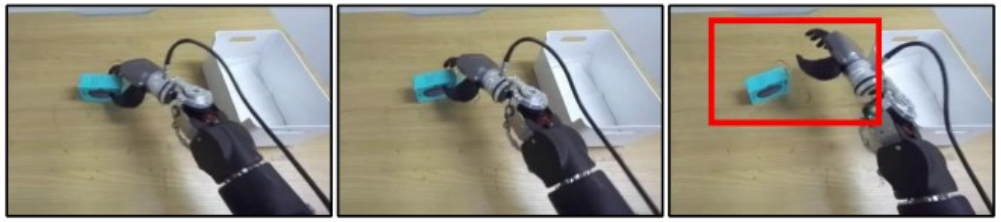
Successful examples shown; overall full-success rates follow later

Analysis ①

Failure modes: human data gives “goals”, robot data gives “execution”

(a) Human data only

Approaches object but fails to grasp: inaccurate wrist/hand pose



(b) Robot data only

Reaches regardless of the target: cannot locate new objects

(c) Shared-action baseline

Generated poses fail to grasp: coordinate alignment misses perception gap

(d) Reverting to robot behavior

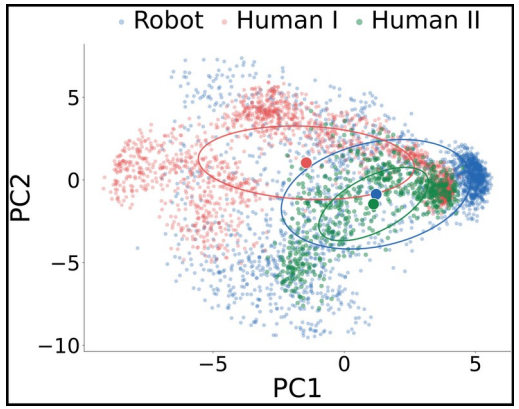
Does seen robot PnP in the air fryer scene instead of opening it

Analysis ②

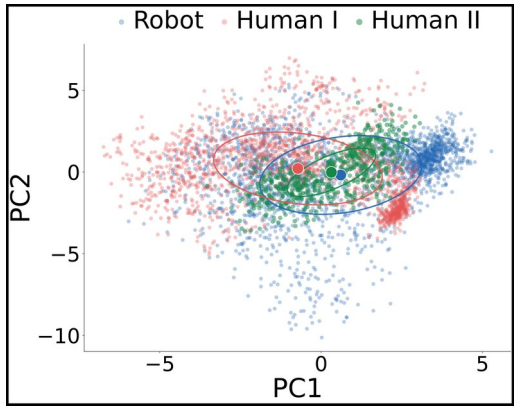
Ablation: tokenizer lifts skill 0 → 49.7; regularizer shrinks latent gap

Action tokenizer	Cross-domain latent reg.	Object transfer	Skill transfer	Latent gap
✗	✗	34.7	0.0	-
✓	✗	43.3	49.7	3.81
✓	✓	**51.8**	**62.7**	**1.53**

- First row = Ego-Pi shared action + hand-geometry alignment
- Latent gap = human–robot latent mean distance ÷ pooled std (Appendix A.5): 3.81 → 1.53, down 59.8%
- Authors: without regularization, robot-conditioned predictions drift toward the robot latent distribution, even for human-taught tasks



Without cross-domain reg.



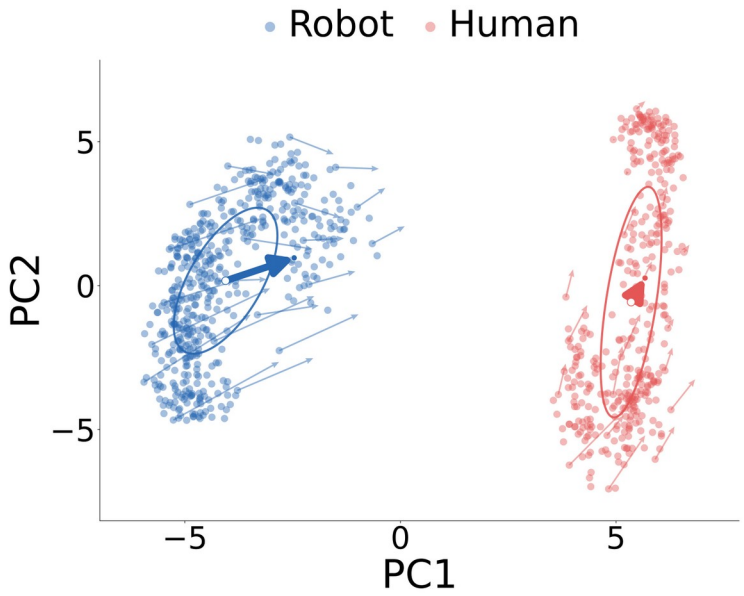
With cross-domain reg.

Tokenizer latent PCA: blue = robot, red = human I (object transfer), green = human II (skill transfer); with regularization, the three means converge

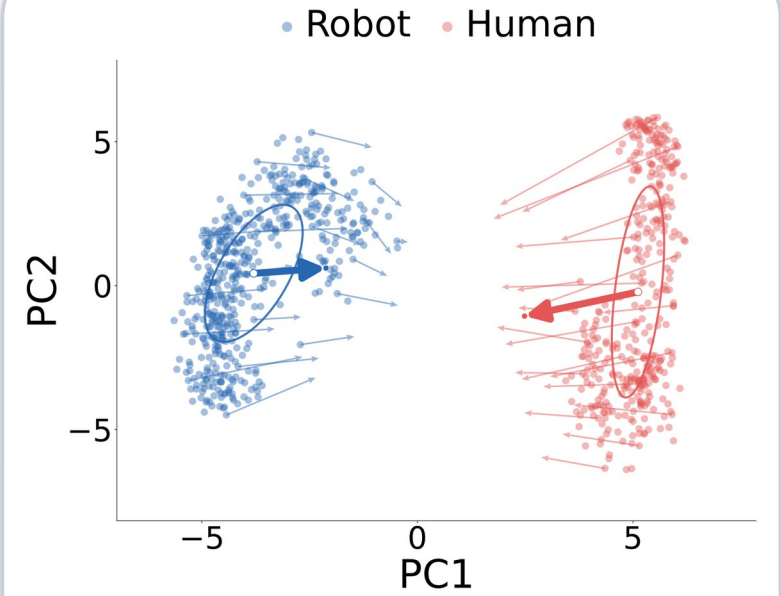
Source: Table 3(a), Figure 9(b), Appendix A.5

Analysis ③

Features: H2RB brings human/robot VLM features closer; gap remains



Hand-geometry alignment only



Full H2RB

Arrows = shift from the adapter

Position change of the same observation before/after the adapter (blue = robot, red = human)

Geometry alignment only

Robot features shift slightly right; human features barely move

Full H2RB

Human features are pulled strongly toward the robot side and the clusters converge, yet remain separate in PCA

Paper: H2RB moves robot-conditioned features toward the human distribution, making both embodiments more compatible in the conditioning space for action prediction.

Source: Figure 9(a), Section 4.2

Analysis ④

Human data speeds motion: wrist ~2.9× faster than robot-demo policy

2.52

cm/s • trained on robot demos

100 teleop demos (50 Open/50 Close)
Task score 65.3

7.36

cm/s • human-demo transfer (H2RB)

50 human demos each Open/Close
Task score 62.7

×2.92

Wrist speed gain

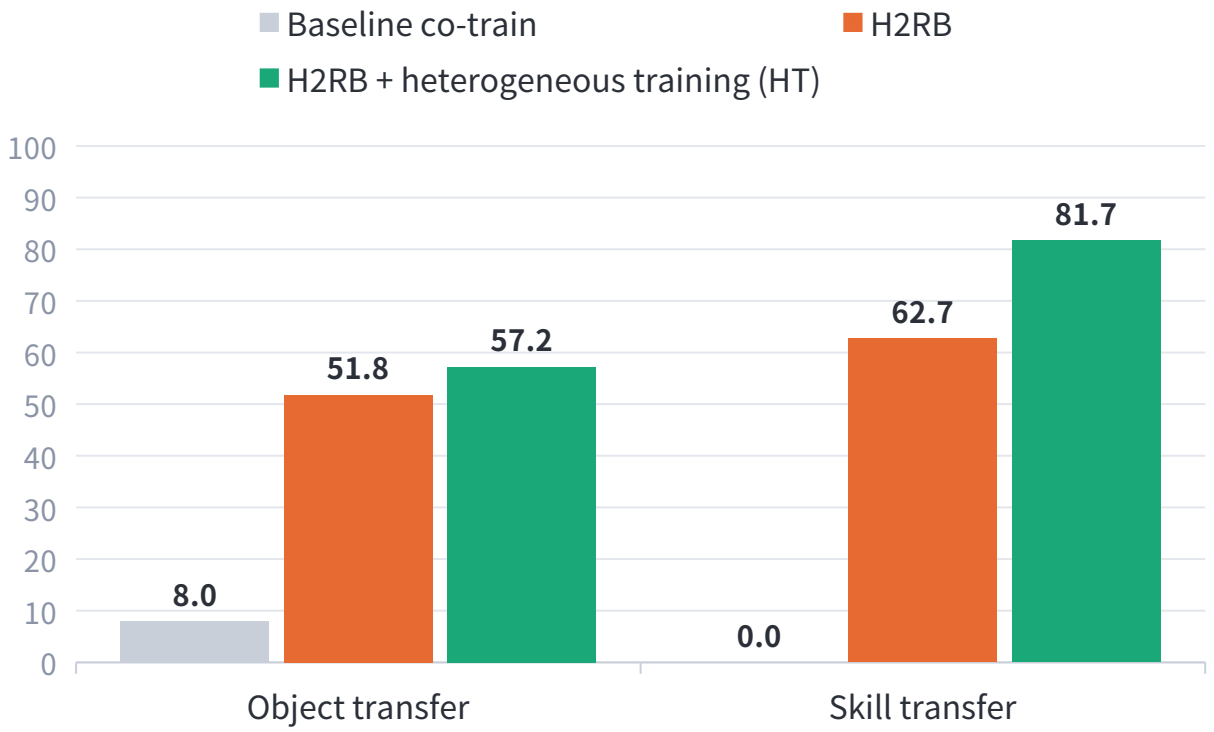
Similar task scores: 62.7 vs 65.3

Authors credit separate decoders for preserving human motion dynamics

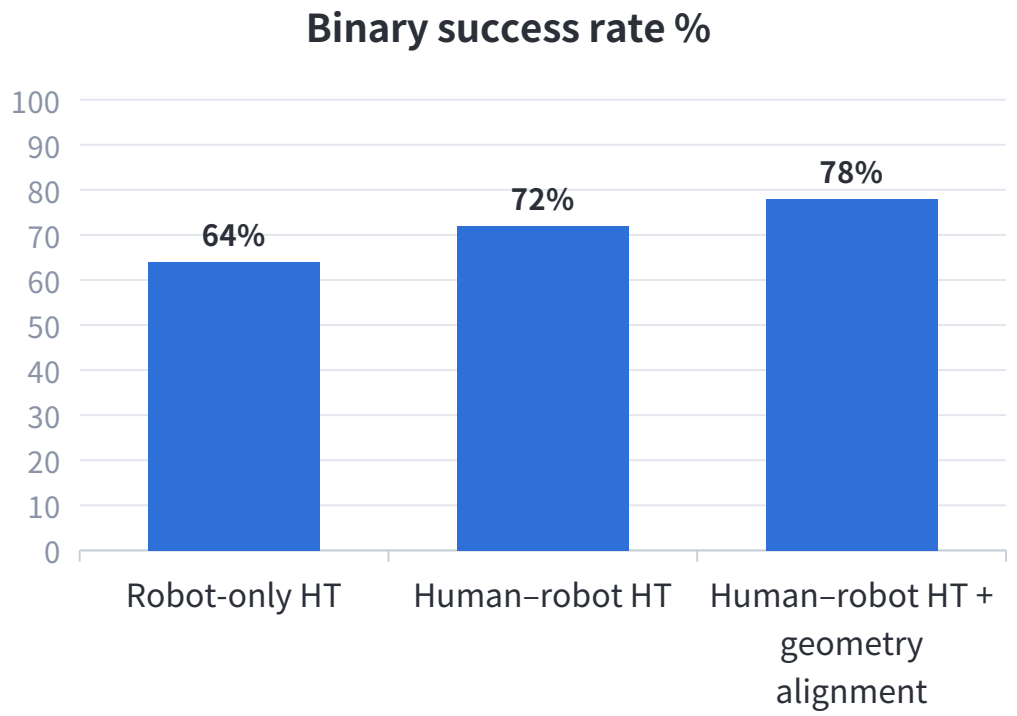
Measurement: wrist speed is computed from consecutive predicted wrist positions at 20 Hz over each successful rollout (including idle phases). The two policies are trained separately, not one policy with swapped decoders. Human demos also avoid the cost of robot teleoperation.

Scaling: large mixed-data training helps; alignment aids other robots

OpenArm transfer (Figure 6)



RB-Y1 parcels, robot-only fine-tune (Fig. 5)



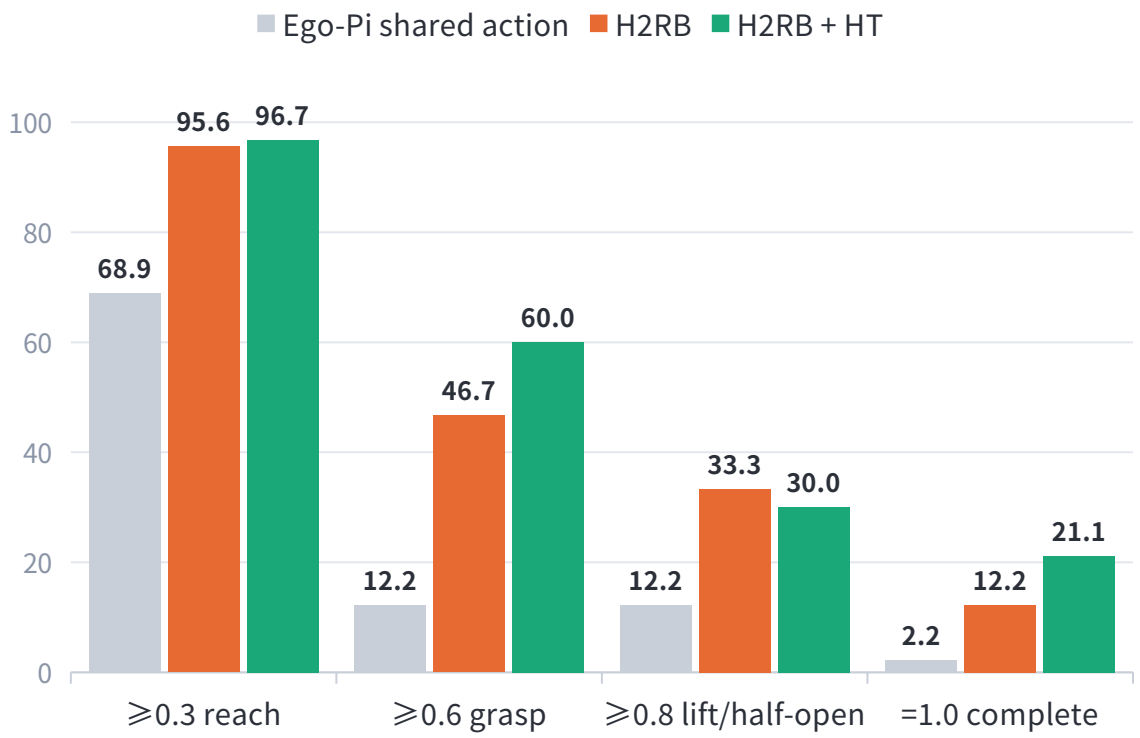
HT: ~14.5k episodes / 5M frames (EgoDex, in-house human data, OpenArm, ALLEX); with instruction re-injection, skill transfer 62.7 → **83.3**

RB-Y1: 20-DoF dexterous hands flip boxes/mailers label-up and push them onto a conveyor; 50 trials; no tokenizer in downstream fine-tuning

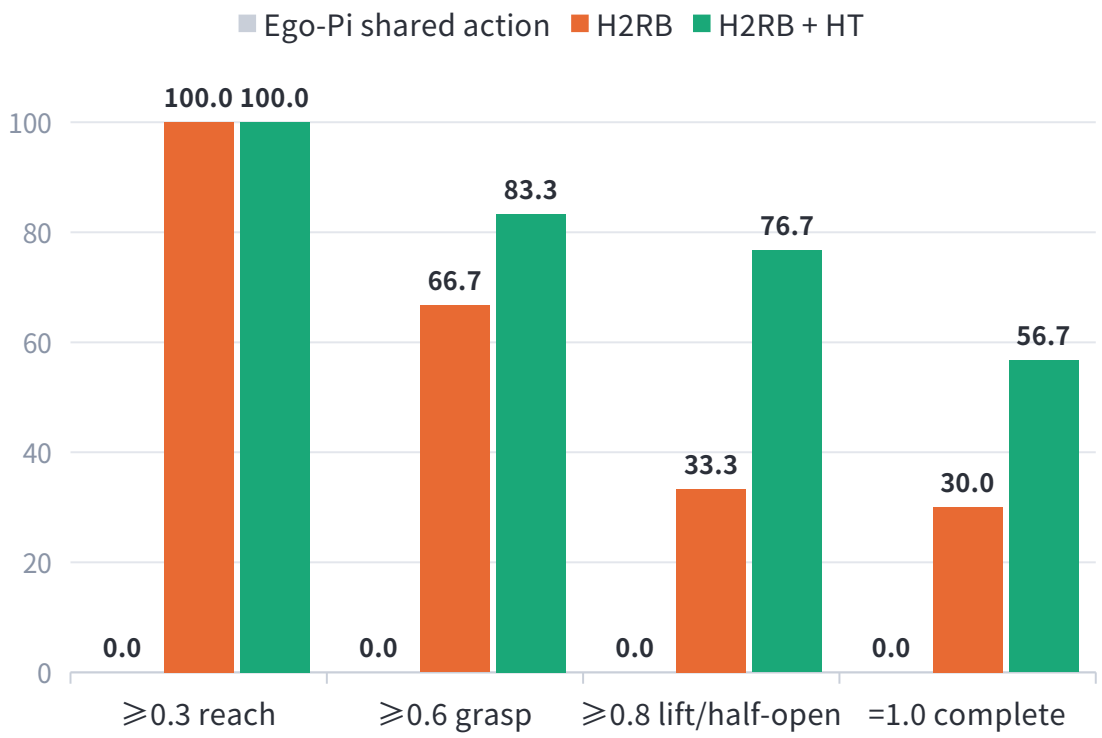
Critical Assessment

Caveat: “reach” and “grasp” drive scores; full success remains low

Object transfer (n = 90), cumulative progress %



Skill transfer (n = 30), cumulative progress %



H2RB full-success rate: object transfer 12.2% (21.1% with HT), skill transfer 30.0% (56.7% with HT). Gains over baselines are real, but reliable deployment is still far off.

Clear and lightweight, but small, controlled evals limit generality

Strengths

Clear problem decomposition

Perception and execution gaps each get a dedicated module, each verified by ablation

Lightweight geometry alignment

Self-labeled frames, no pairing or image synthesis; usable for heterogeneous training and other robots

"Shared skills, separate execution" works

Skill transfer 0 → 62.7, while keeping the faster motion of human demos

Limitations / Caveats

Small sample sizes

One policy per condition, no multiple seeds; only 30 rollouts for skill transfer

Highly controlled human data

Same environment, similar viewpoints, same placement protocol; tokenizer trained per setting

Alignment is imperfect

Deployment requires the human decoder; authors admit the regularizer does not ensure sample-level correspondence

Relies on reconstruction pipeline; reporting inconsistencies

HaWoR → retargeting → IK errors unanalyzed; Table 1/2 captions mismatch their contents

Summary • Take-aways

Human-to-robot: align representations, not identical actions

01 Unified action space: only step one

Without closing the perception and execution gaps, shared-action baselines still score 0 on new skills.

02 Bridges: geometry + split decoders

Hand geometry brings VLM features closer; the shared latent space handles “what”, separate decoders handle “how”.

03 Strong gains, not yet reliable

Object 51.8 / skill 62.7 (57.2 / 81.7 with HT); full success 12%–57%; small-scale evaluation.

04 Human video: scalable supervision

Next: validate on more in-the-wild human data, more random seeds and more tasks.